

A Multidimensional Instrumental Characterisation of Dairy vs. Plant-Based Cheese Analogues: Integrating

Rheology, Tribology, and Surface Profilometry to Predict Oral Texture

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BACKGROUND



Sensory Panel Limitations [1]

- Biological variation between individuals persists even after training
- Costly & time-consuming to recruit and maintain



Growing Vegan cheese Market

- Growing awareness of lactose intolerance and environmental & ethical issues is shaping consumer choices [2]
- Texture & mouthfeel remain key barriers to consumer acceptance [3]

Aim: To investigate whether a synthetic tongue tribometer can provide reproducible, objective measurements of cheese lubrication and mouthfeel

METHODS

01 Surface Topography & Rheology

Cheese samples: Mature cheddar (dairy) vs. Commercial vegan analogue.

- Rheometer used for temperature sweep 25°C → 75°C
- G', G'' and $\tan \delta$ measured to characterise viscoelastic behaviour
- Alicona used to measure surface roughness (Sa, Sq) of both cheese types

02 Tongue Design & Fabrication

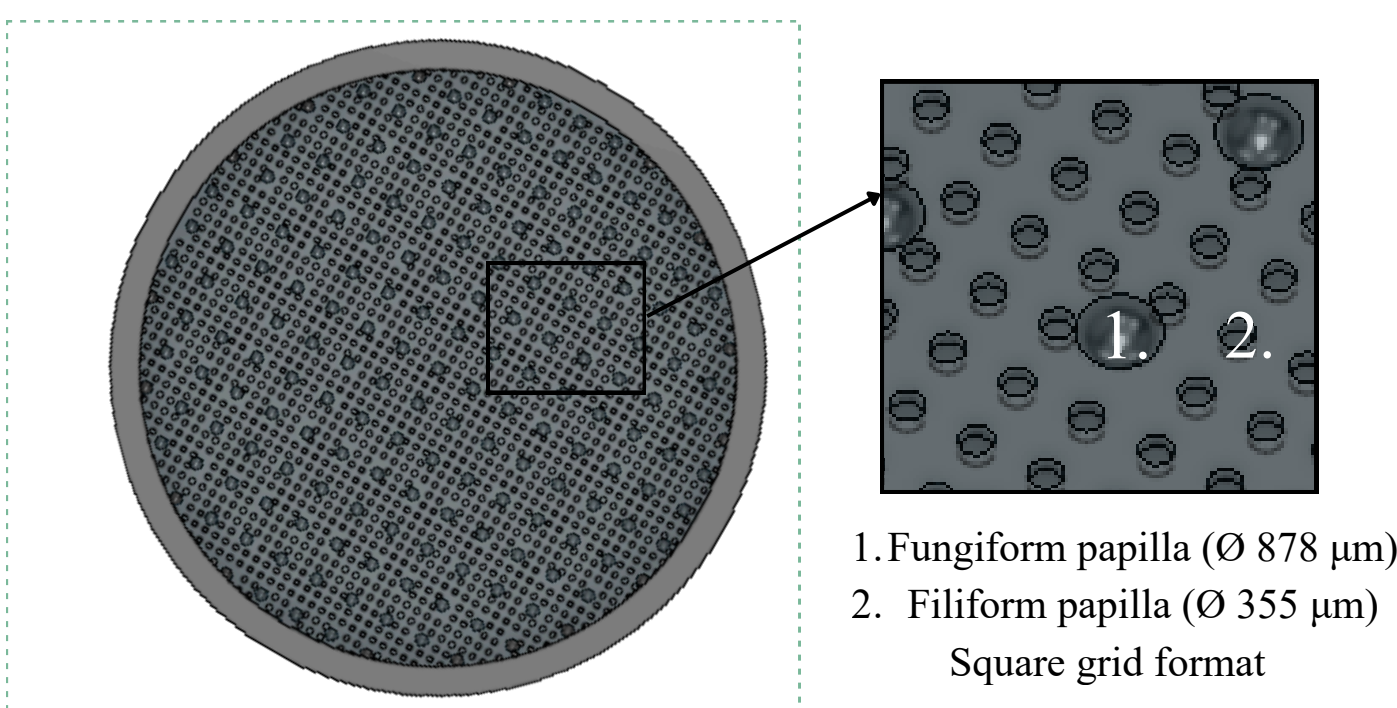


Fig.1: CAD mould design of synthetic tongue surface

Synthetic tongue fabrication:

1:1 Ecoflex : Span 80 (0.5 wt%) → cured 5 hrs

03 Tribology Set Up:

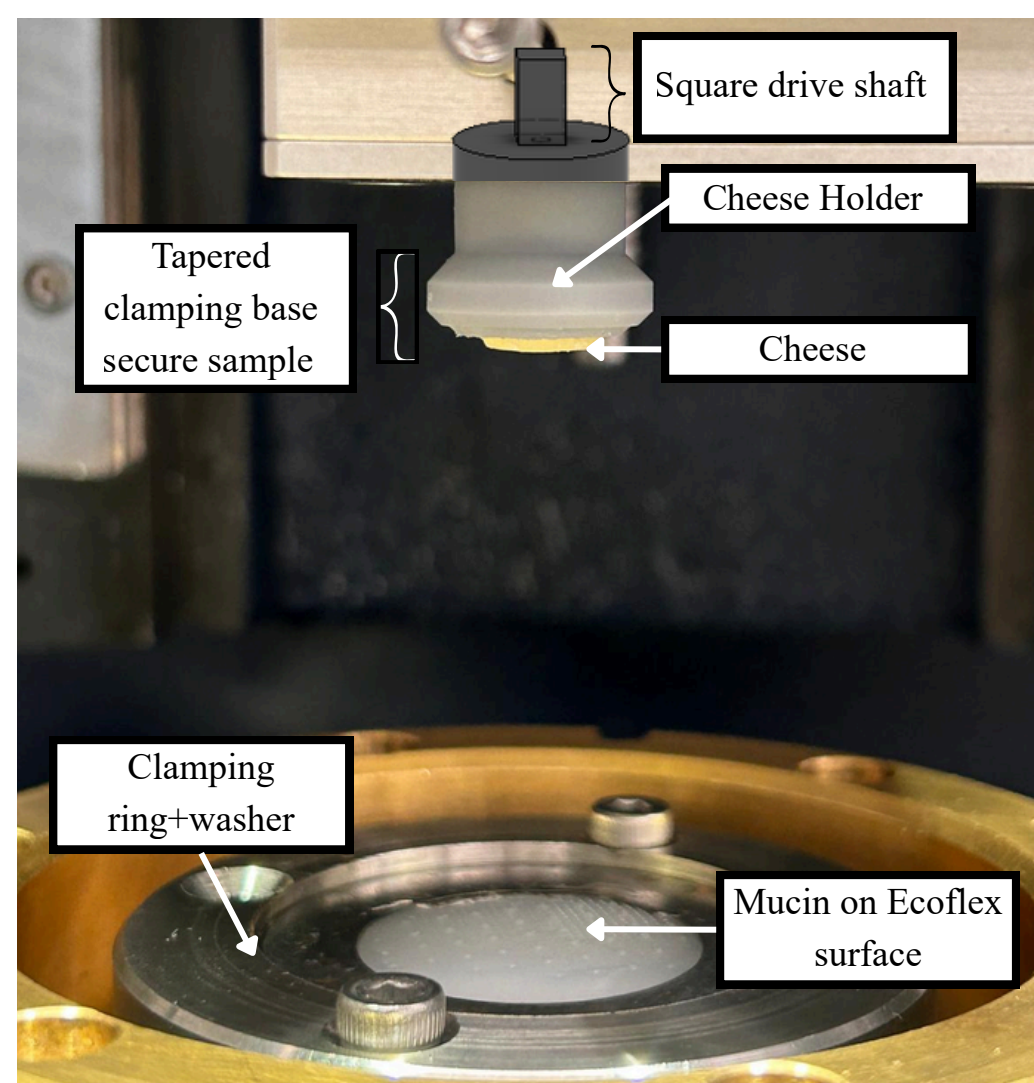


Fig 2. Tribology experimental setup

Oral Tribology testing:
Mucin-lubricated tribology
(0.5–30 mm/s)
→ Stribeck curves

OBJECTIVES

- Assess surface topography and roughness of both cheese types using Alicona 3D profilometry (Sa, Sq)
- Characterise the temperature-dependent viscoelastic behaviour of dairy and plant-based cheese (25-75°C) using dynamic rheological measurements
- Evaluate tribological behaviour of cheese samples using a synthetic tongue tribometer, correlating CoF with smoothness and creaminess perception

04 Tribology Data Processing:

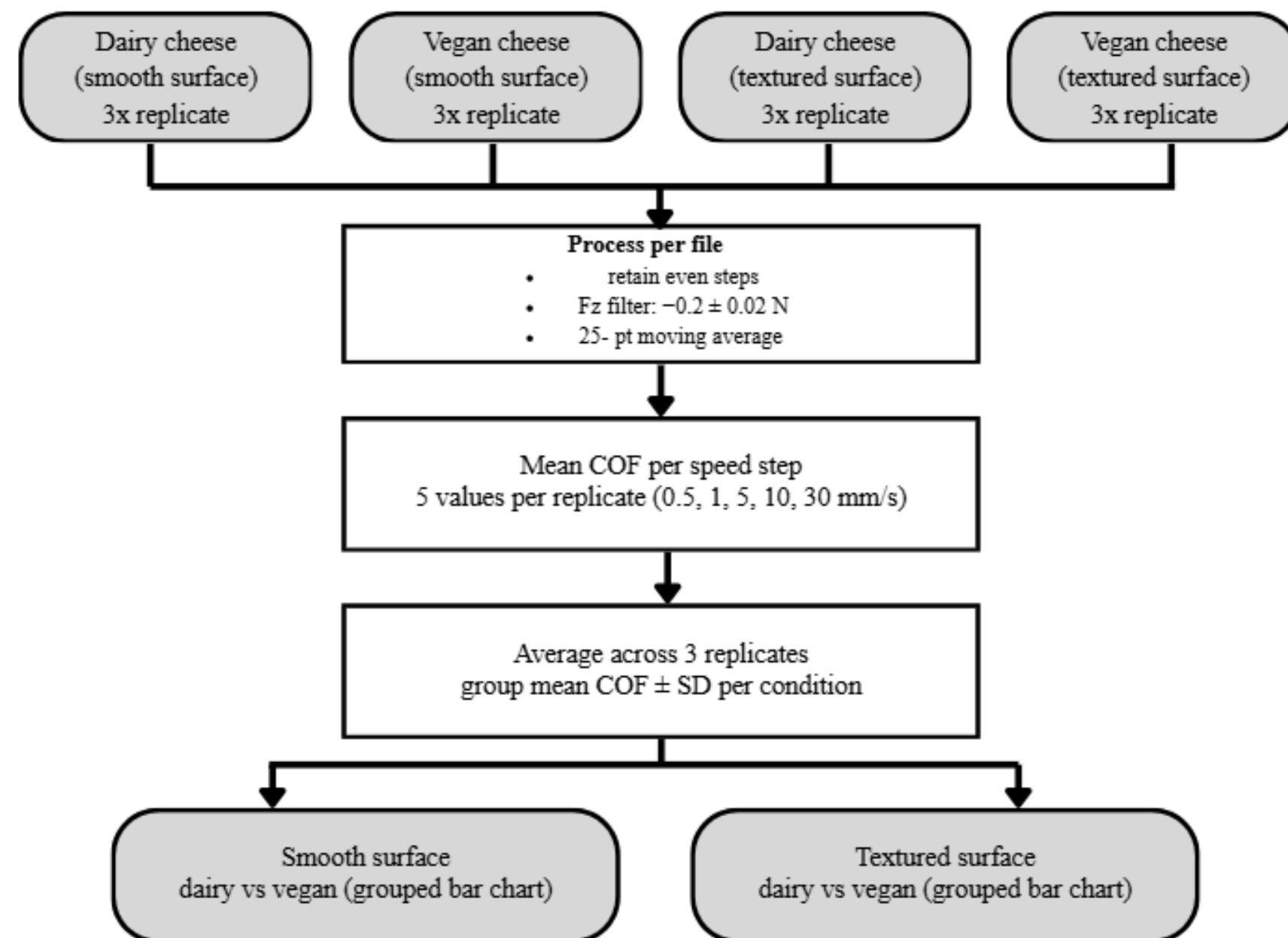


Fig 3. Tribology data processing workflow on MATLAB

Mean CoF ± SD across 3 replicates → 4 conditions (dairy/vegan × smooth/textured)

RESULTS

01 Surface Topography

Parameter (μm)	Dairy cheese	Vegan Cheese
Sa	3.94 ± 0.78	2.04 ± 0.31
Sq	5.18 ± 1.12	2.64 ± 0.44

Table 1: Surface roughness parameters (Sa, Sq) of dairy and vegan cheese (Sourced from Marks & Spencer)

Key finding:

Higher roughness in dairy may suggest greater microstructural complexity → potential influence on tribological behaviour and oral texture perception [4]

02 Rheology

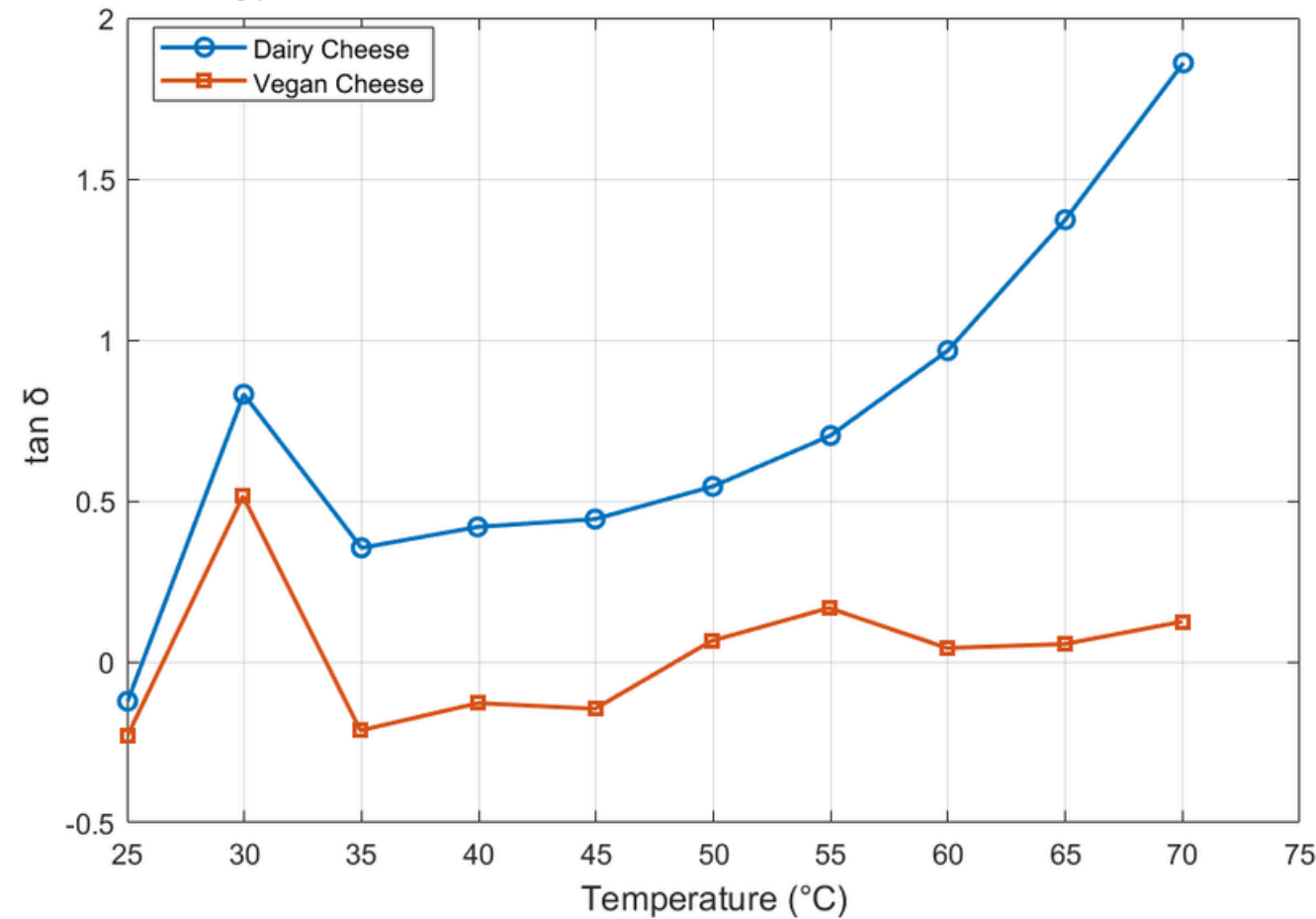


Fig 4. $\tan \delta$ vs temperature (25–70°C) — dairy vs vegan cheese

Key finding:

- Dairy: Structural breakdown of casein matrix ($\tan \delta > 1$ at ~30°C) suggests structural transition enabling fat release[2,4]
- Vegan: Starch-hydrocolloid network maintains solid dominance ($\tan \delta < 0.2$), likely limiting fat release [2,5]

03 Tribology Testing

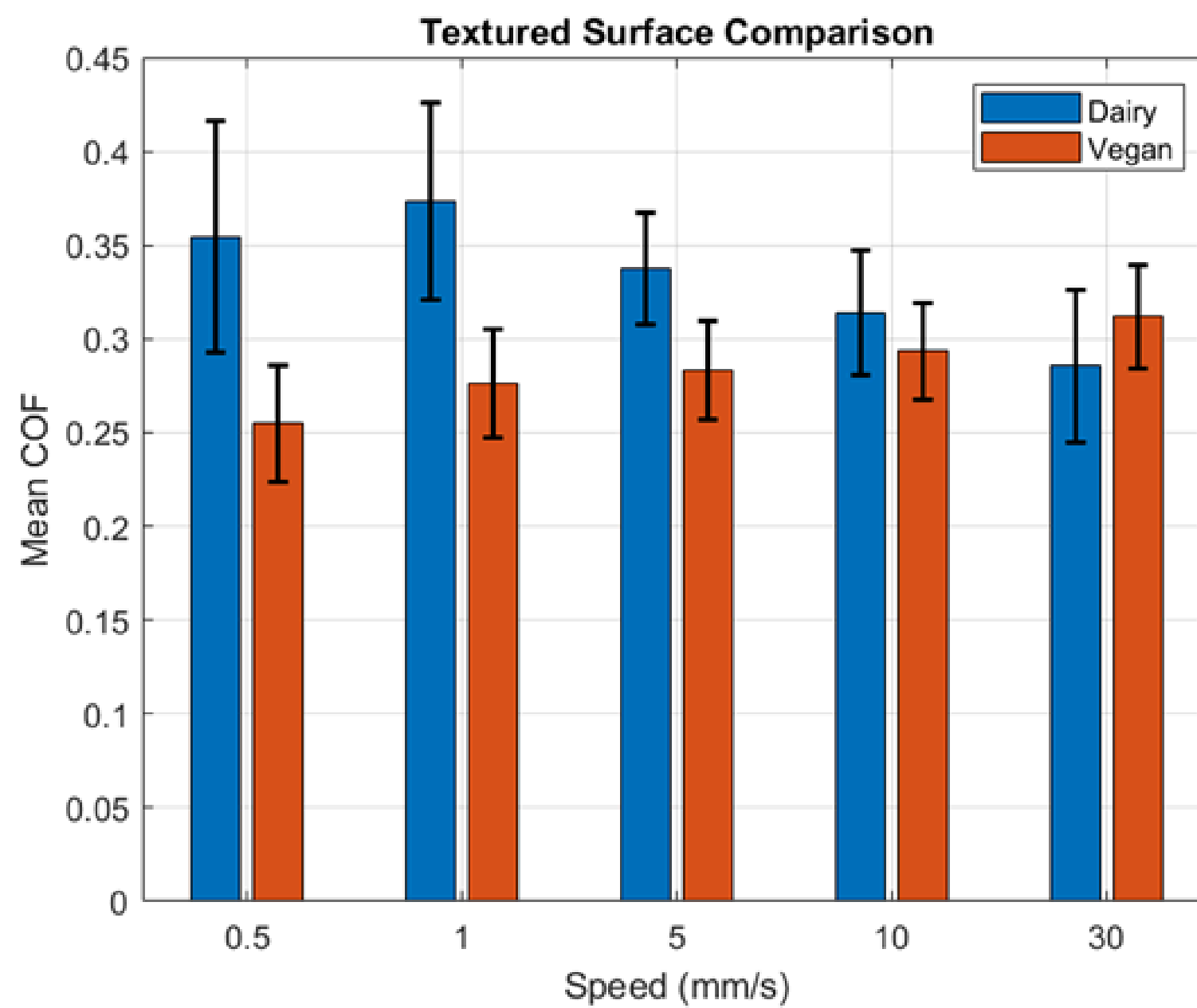


Fig 5. Stribeck curves: dairy vs vegan cheese on textured surface (mean CoF ± SD, n=3)

Key finding:

- Dairy CoF decrease with speed may suggest fat lubrication only activates above a critical sliding speed [6]
- Vegan rigidity promotes fracture over flow, fragmentation likely disrupts the mucin film, increasing friction with speed[5,7]
- Papillae are the key mechanism → smooth surface results suggests that tongue geometry, not cheese composition alone, drives lubrication differences [1]

LIMITATIONS & FUTURE WORK

01 Limitations

- Room temperature testing — 37°C may better replicate oral conditions
- Porcine mucin — doesn't fully replicate human saliva complexity
- Grid-based papillae — doesn't reflect natural non-uniform tongue architecture
- Anomalous vegan $\tan \delta$ — possibly plate slippage or instrument sensitivity

02 Future Work

- Temperature-controlled tribology at 37°C
- Human saliva or more complex simulant
- Papillae redesign reflecting natural spatial organisation
- Repeat vegan rheology with clamped geometry to confirm anomalous values
- Replicate mechanical compression testing to validate $E = 14.653$ kPa

04 Validation Tests For Synthetic Surface

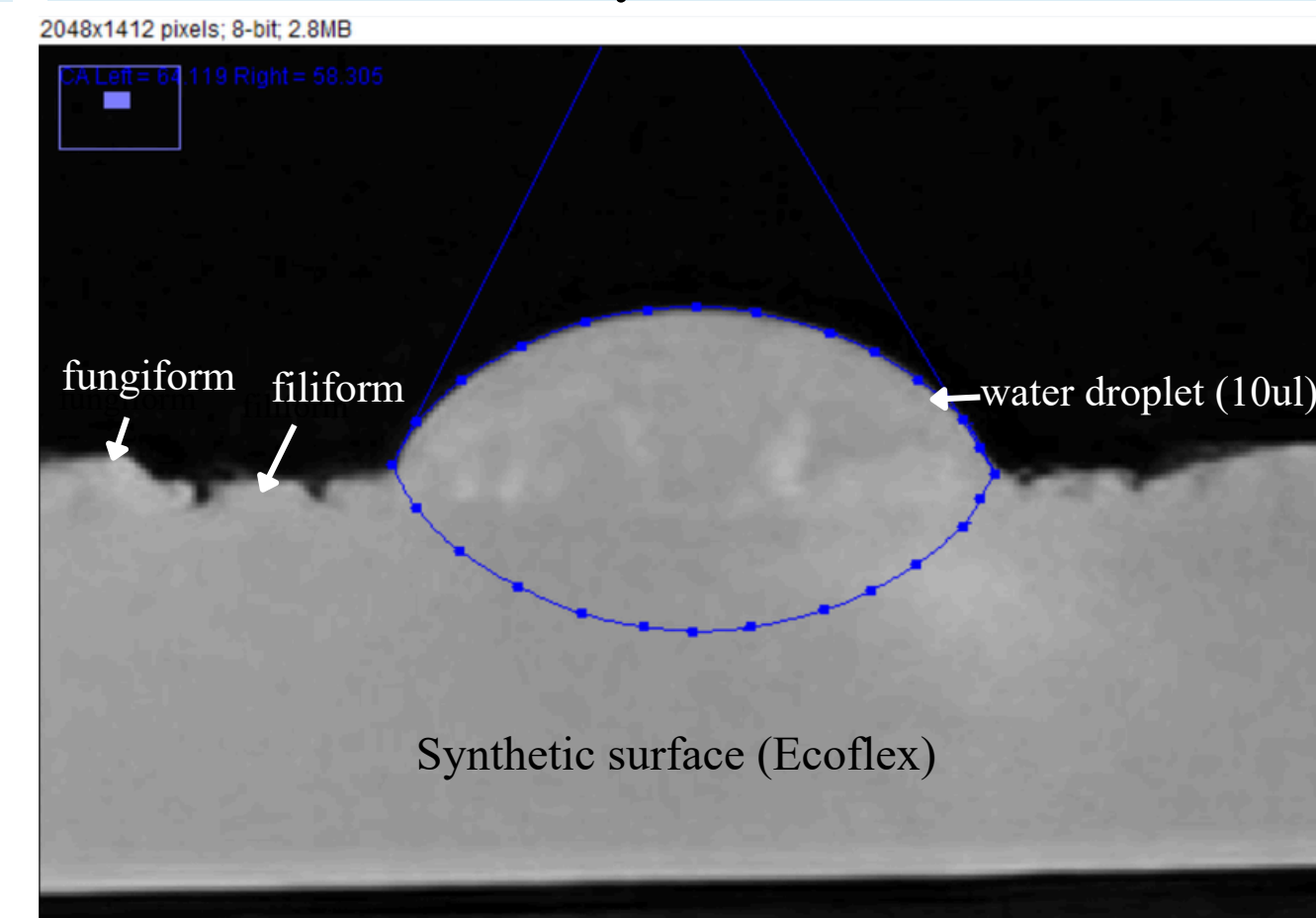


Fig 6. Contact angle of EcoFlex synthetic surface

Contact angle: $58.6 \pm 0.7^\circ$, consistent with literature [1]

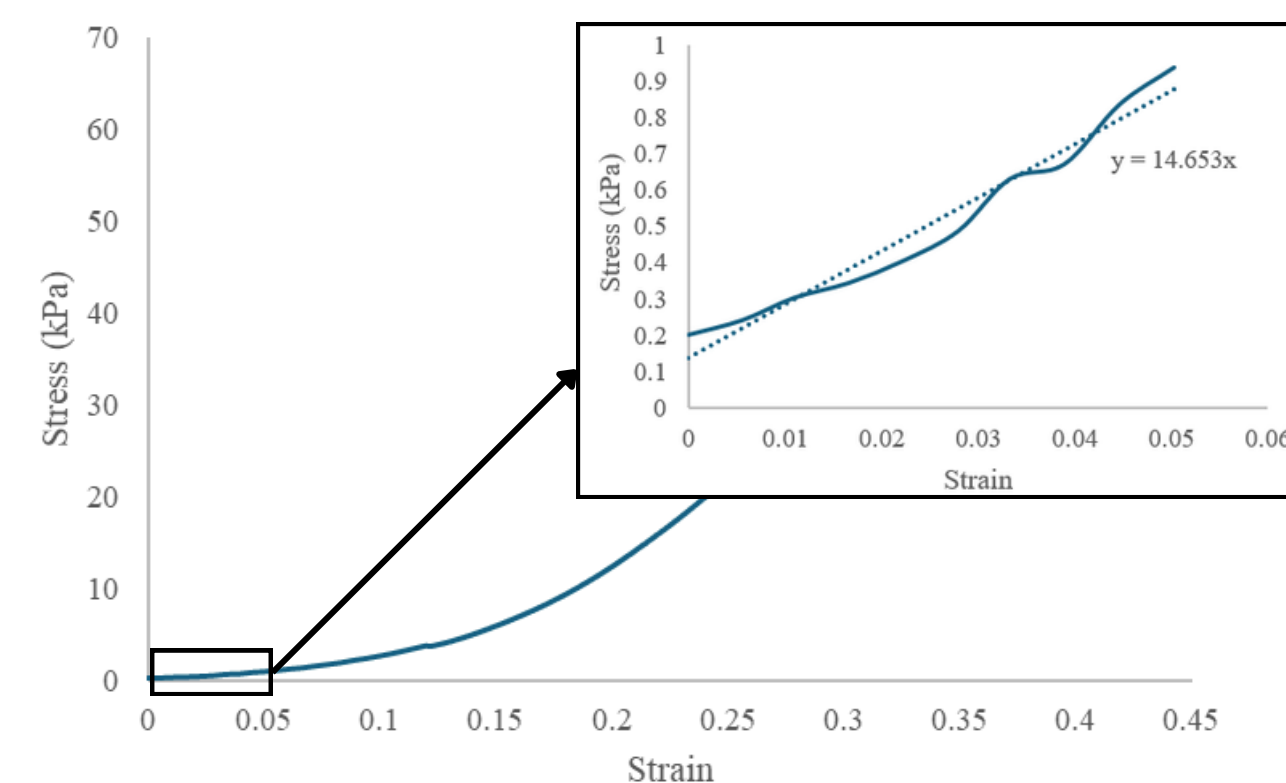


Fig 7. Stress-strain curve of EcoFlex synthetic tongue surface — linear region (0-0.05 strain) used to calculate Young's modulus

$E = 14.653$ kPa — closer to real tongue tissue (pig tongue < 5 kPa) than conventional PDMS (2.4 MPa) [1]

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